Mechanism Is Not Meaning:
A Verification Methodology for Cognitive-Architecture Prototypes

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Abstract

Building a cognitive-architecture prototype invites a recurring confusion between two claims that resemble each other: that a mechanism is correct, and that it carries meaning. The pressure to overclaim is strongest when a system can be graded on inputs chosen to make it succeed. This paper presents a verification methodology that holds the two apart, demonstrated as a case study on Psyche, a dependency-free substrate assembled into a closed cognitive cycle. The methodology combines four practices: a separation of duties between building and certifying, loop-closure driven by system-generated input, adversarial and property-based testing, and pre-registration with anti-circularity gates. Under it, the substrate's properties are proven and the cycle is verified as a mechanism, with no cognitive claim. The methodology is then exercised on three attempts at an externally graded cognitive result: semantic priming, the fan effect, and an interoceptive modulation. Each was halted before any graded number, for a structurally distinct reason. The recurring obstacle, the relationship between the mechanism and an external yardstick rather than the mechanism itself, is itself the finding. The meaning bar was not cleared; the results concern method and mechanism, not cognition.

1 Introduction

A system built toward cognitive claims is under steady pressure to be described as more than it is. Each working mechanism invites a larger reading, and the least demanding way to support that reading is to grade the system on inputs chosen to make it succeed. The effect is a recurring confusion between two statements that resemble each other and are not the same: that a mechanism is correct, and that it carries meaning. Holding them apart is the central difficulty in this kind of work, and it is a methodological problem before it is a scientific one. This paper is about that problem. Its contribution is a verification methodology for building cognitive-architecture prototypes without the overclaiming, and its evidence is a case study in which the methodology was carried, from the substrate upward, through one such prototype.

The prototype is Psyche.[1] It is a dependency-free substrate, a graph store and a spreading-activation engine, assembled into a closed cognitive cycle. It was developed under a longer-term motivation to explore biologically-motivated, non-LLM cognitive architectures, but that motivation is not what this paper evaluates; the paper reports only the methodology used in building toward it. Two grades of evidence are kept apart throughout. The substrate's properties are proven, in a precise sense established later, to be deterministic across separate processes, internally consistent through every change, and free of any read that alters what is stored. The cycle built above the

[1] Psyche (developed in the repository under the working name Nous).

substrate is verified as a mechanism and nothing more. It is not claimed to perceive, to attend, or to mean. Naming it a cognitive cycle borrows a control structure from an established tradition; it does not assert that the structure performs cognition.

The methodology that governed this construction is four practices. The first is a separation of duties, so that the role which writes a mechanism is never the role that certifies it. The second is loop-closure, driving each stage with the input the previous stage actually produced rather than input placed by hand. The third is adversarial and property-based testing, which asserts invariants designed to break a claim rather than to confirm it. The fourth is pre-registration with anti-circularity gates, fixing a prediction and the direction it must take before any graded number is computed. None of the four is novel on its own. Their combination, turned on a developer's claims about a system they are building, is the contribution this paper makes.

The discipline is tested where the temptation to overclaim is strongest: at the step from a working mechanism to an externally graded cognitive result. Three times the project reached for such a result against data it did not author, attempting semantic priming, the fan effect, and an interoceptive modulation of the cycle. Three times the anti-circularity gates halted the attempt before any graded number existed, and each time for a structurally distinct reason. One claim could not be separated from the construction of the substrate. One named a generating mechanism the engine does not contain. One rested on an external signal that does not vary on the work the system does. That three independent attempts halted at the same place, the join between the mechanism and an external yardstick rather than the mechanism itself, is itself a finding. The meaning bar was not cleared, and this paper does not present it as cleared. It presents a result about method and about mechanism: a discipline that caught its own premature claims before they became numbers, over a substrate and a cycle that remained verified throughout.

The paper proceeds as follows. Section 2 describes the substrate and the closed cycle at the level of mechanism. Section 3 sets out the methodology and the structural errors it caught while the system was built. Section 4 recounts the three attempts at a graded cognitive result and the reason each was refused. Section 5 gathers the three into the observation that the recurring obstacle is the external yardstick, not the mechanism. Section 6 states the limitations, and Section 7 positions the methodology and the prototype against prior work. The verification counts and the data behind the three attempts are collected in the appendices.

2 The Substrate and the Closed Cognitive Cycle

Psyche is built in two layers that carry two different grades of evidence, and the distinction between them is maintained throughout this paper. The lower layer, the *substrate*, is a graph store together with a spreading-activation engine; its properties are proven, in the precise sense established below: identical output across separate processes under randomized hashing, mutual consistency of the graph indices after every mutation, and reads that never alter stored state. The upper layer assembles the substrate into a single loop, perceiving, activating, clustering, selecting, pruning, and writing, whose stages are verified as mechanisms but make no claim to reproduce a cognitive phenomenon. We call the loop a *cognitive cycle* because its shape is taken from the global-workspace tradition [Baars, 1988, Franklin et al., 2014], as a structural template rather than a theory the system is claimed to instantiate; the term names a control structure, and its use here asserts nothing about cognition. What the loop is shown to do is to *close*, as a mechanism, and that is a narrower and more exact statement than it may first appear.

2.1 The substrate store

The store is a directed graph whose vertices are concept *nodes* and whose edges are weighted *roads*. Nodes hold no state of their own; everything that varies lives on the roads. A road carries a scalar *strength* together with a record of how it has been used, advances through a small set of discrete quality tiers as it is traversed, and decays with elapsed time when it is not. A road may also be *closed*: a closed road remains in the graph and stays traversable by a sufficiently strong cue, so it is leaky rather than severed, but it no longer conducts ordinary activation.

The load-bearing decision in the store is that reading and writing are separated at the level of decay. An earlier arrangement applied decay at the moment of access and wrote the decayed value back, with two consequences a casual reading would miss: every read mutated the store, and, because that same step re-evaluated closure, a sequence of reads spread over time could let a closed road drift back open although nothing had ever crossed it. Psyche instead splits decay into a pure view and a persisting write. A read returns a decayed *copy*, computed from the elapsed time but stored nowhere, so reading changes nothing; only the active operations, traversal and reinforcement and the offline maintenance passes, persist a decayed value. Closure is made *sticky* by the same principle: it is re-evaluated only on an active qualifying crossing, never by elapsed time and never by a read, so no amount of passive inspection can reopen what only an event should reopen. These are the properties the substrate is proven to hold: reads are pure, closure does not drift, the graph indices stay mutually consistent through every insertion and deletion (including on adversarially constructed graphs), and a deletion removes a node completely, leaving no dangling entry in any index. The corresponding assertion counts appear in Appendix A.

2.2 The spreading-activation engine

Activation is the substrate's one dynamic operation. Given a set of seed nodes, the engine propagates a diminishing quantity outward along the roads and reports how much activation each reached node accumulates. The mechanism is the spreading activation of ACT-R [Anderson and Lebiere, 1998]; Psyche borrows it without that architecture's empirical commitments. Two design choices make the computation well defined.

The first is bounded single-expansion. When activation arriving along several paths converges on a node, the quantities must add; the hazard is that the node, once re-expanded as later paths raise its value, propagates the surplus, and everything downstream of the convergence point is counted more than once. The error is the residue of a shortest-path intuition, in which the first time a node is settled is final, carried into a setting where contributions accumulate rather than compete. Psyche expands each node exactly once, at the first hop at which it is reached, carrying the full sum of all paths of length up to that hop; activation that converges by equal-length paths is therefore summed once and exactly, independent of the relative strengths of the converging roads.

The second choice secures determinism. If expansion follows an activation-ordered frontier, which late contribution is discarded depends on incidental orderings of road strength. Separately, if same-hop siblings expand in the order a set of machine-generated identifiers happens to iterate, that order is randomized from one process to the next by the hashing of those identifiers, and the result moves between runs for no reason present in the graph. Psyche instead expands hop by hop and freezes each layer before it acts: all contributions that reach distance k are accumulated into a snapshot, the snapshot is frozen, and only then do the hop-k nodes expand, reading their values from the frozen snapshot rather than from one another. Because no node within a layer can observe another's expansion, the outcome cannot depend on the order in which nodes or their outgoing roads are visited. The engine is in consequence deterministic across separate processes

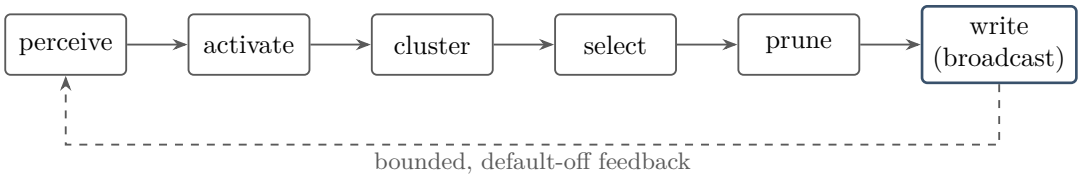


Figure 1: The closed cycle of Psyche as information flow at the level of mechanism. A cycle proceeds from perception to activation, clusters the perceived sources, selects a single winning coalition, prunes the region that coalition reaches across a bridge, and performs one bounded write, the broadcast, that biases the next cycle. The figure depicts graph operations and a single bounded write; the verbs are select, prune, and write, not decide, attend, or experience. The feedback edge is bounded, default-off feedback. No cognitive process is depicted, and no cognitive claim is made.

under randomized process hash seeds (5–8 per suite, in fresh interpreters), and, like the store, it computes as a pure read: a spread mutates nothing.

Together the store and the engine form the substrate, the proven layer of the system. Their two suites contribute 45 and 35 assertions respectively, 80 in all, within a verification suite that currently stands at 20 suites and 1,372 assertions with no failures (Appendix A). Everything above this layer is built on it and claims less.

2.3 The closed cognitive cycle

Above the substrate, Psyche arranges the operations of the store and the engine into a single loop, shown in Figure 1. A cycle begins by *perceiving*: the input text is matched against the substrate, and the nodes it names become the cycle's sources. This step is what makes the rest of the loop a genuine test rather than a staged one, because the sources are produced by the system from its own contents rather than placed by hand; the loop is driven by input it generates, not by input chosen to make it succeed.

Perception rarely yields a clean set of well-separated sources; it names several that lie close together in the graph, and an activation field built directly on such a set produces basins that overlap and collapse rather than separate concerns. The first stage after perception therefore groups the sources before any field is constructed. Psyche builds a small proximity graph over the perceived sources, using only the strength of the roads between them, merges them by single linkage [Sibson, 1973], and seeds the downstream field from one representative per group. The rationale is that the grouping must come from the structure of the substrate, not from the activation it is about to compute, so that the later stages are not handed a partition that already encodes the answer; using road strength alone keeps the pre-stage topological. The merge is a deterministic, pure, integrity-preserving operation, and it is verified as such: it asserts nothing about meaning, only about which sources the graph places together. The threshold that governs how close is close enough is a fixed design constant rather than a calibrated one, and we mark it as such instead of presenting it as tuned.

From the grouped sources the engine computes its field, and the co-activated region is partitioned into candidate coalitions, of which exactly one is selected as the cycle's result. Selection is a competition normalized for coalition size, so that a coalition does not prevail merely by being large, and it returns a single winner deterministically. Selection then prunes from the winner any region it has reached across a bridge: a perceived concern can extend, through a single connecting road, into a neighbouring region the input never named, and that unnamed region should not enter the result on the strength of one bridge. The pruning rule is parameter-free. A reached element is kept only if

it is attached to the named region at least as strongly as to the unnamed one; where attachment is stronger toward the unnamed side, the rule finds a cohesion valley, a seam of relatively weak connection, and cuts there. The rule has one corner case, which we state rather than conceal: an unnamed region bridged into the named one at least as strongly as it coheres internally presents no valley to cut at, and is retained. The divergence is one-directional and on the safe side, in that the rule may keep an unnamed region a stricter criterion would drop but never prunes a member the input actually named. Selection, pruning included, is a pure read; it is verified as a mechanism and asserts nothing about cognition.

Up to this point the cycle has only read the substrate. Its one write is the final stage, the broadcast: the winning coalition is fed back onto the substrate by reinforcing its internal roads, biasing the next cycle toward the same region. This single write is what turns a pipeline into a loop, and it is also the first opportunity the substrate has to corrupt itself over repeated cycles, so its boundedness is established in both directions. The reinforcement reuses the store's ordinary strengthening rule, which adds to a road's strength a fixed fraction of the distance that remains to its ceiling,

$$s \;\leftarrow\; s + \gamma\,(1 - s), \qquad 0 < \gamma \leq 1,$$

so that each application closes part of the gap to $s = 1$ without ever reaching or crossing it. Boundedness is thus a structural consequence of the update rather than a limit imposed afterwards: the broadcast contains no cap, no ceiling test, and no tunable magnitude of its own, the gain γ being the store's existing default. In the other direction, the feedback does not lock the loop onto its first winner; a sufficiently strong later input still moves the result, which was confirmed by driving one region to dominance over sustained reinforcement and then verifying that a competing input takes the workspace, with the first region checked to be dominant immediately before it yields. The broadcast is off by default, and with it off the loop is a pure read whose output is byte-for-byte identical to the pre-broadcast pipeline; enabling it adds exactly one bounded write per cycle and nothing else.

With the broadcast in place the loop is closed: perception, activation, clustering, selection, pruning, and the single feedback write run end to end on input the system generates, the read portion is deterministic across processes and pure, and the write portion is the one bounded reinforcement just described. This is the precise content of the claim that the cycle closes. It is a statement about a mechanism, a pipeline that runs to completion on its own output, repeatably and with no side effect beyond one bounded write. It is not a statement that the loop perceives, attends, decides, or means anything. Each of its stages, the clustering, the competition and pruning, and the bounded broadcast, has been independently and adversarially re-verified for the same properties the substrate is held to, namely determinism across randomized process hash seeds, preserved integrity, and read-purity, and for none of them is any cognitive claim entered; their assertion counts are recorded in Appendix A. The loop is built and its mechanism is verified. Whether anything it computes corresponds to a cognitive phenomenon is a separate question, taken up, and left open, later in this paper.

3 The Verification Methodology

The substrate and the cycle of the previous section are the case study; the contribution of this paper is the discipline that determined what could be claimed about them, and at what grade of evidence. That discipline exists to counter one recurring hazard. A project that builds toward cognitive claims is under steady pressure to read more into a working mechanism than the mechanism has earned, and the easiest way to do so is to grade the system on inputs chosen to make it succeed. The

methodology is four practices that, taken together, force every claim to earn its tier and hold two statements apart: that a mechanism is correct, and that it carries meaning. The first is settled against hard properties, the second against external data. The practices are arranged so that neither test can quietly substitute for the other.

Two qualifications belong at the front, not only in the limitations. First, the four roles described below are AI subagents operating under enforced lane rules, within a single project directed by one person. They are a separation of roles and contexts, not independent human teams and not external auditors. Second, the discipline is reported as what it did, not as a guarantee. It caught the structural errors recounted below and it stopped three premature claims, but it is demonstrated on one system, and it establishes neither freedom from bugs nor that it would transfer to another project.

3.1 Separation of duties

The work is divided among four roles, and the division is the point. An architect role specifies each component together with the external criterion by which it will be judged, and maintains the project's record of what is built against what is only designed. A builder role implements each specification test-first. A separate tester role then re-verifies the result: it does not trust the builder's tests, it writes its own hostile and loop-closed suites, and it never edits production code. A validator role enters only when a cognitive claim is at stake, and grades that claim against external data under a prediction fixed in advance. The tester's posture is adversarial, in that its suites are built to break the builder's claim rather than to confirm it; the validator's posture is the same, turned on claims of meaning.

The reason for the division is that the role which writes a mechanism should not be the role that certifies it. When author and certifier coincide, verification decays into self-attestation: the tests come to encode the author's own model of the code, and they pass for the same reason the code was written. Holding the tester apart makes agreement between builder and tester informative, because the tester was trying to falsify rather than to agree. The same logic puts meaning beyond the reach of both: a cognitive claim is handed to the validator and graded against data the project did not author, since a mechanism's own developers are not placed to certify what it means. The tester's checks are largely property-based. Rather than fix one output for one input, they assert invariants that must hold across inputs of a kind, such as determinism across processes, purity of reads, and preservation of graph integrity, in the tradition of property-based testing [Claessen and Hughes, 2000, MacIver and Hatfield-Dodds, 2019]. These re-verification suites are separate from the builder's, and are tallied in Appendix A.

These roles are AI subagents under enforced lane rules, not human teams, and the distinction is not a quibble. The separation is one of role and context, an architecture of constraints on who may do what, with the tester and validator confined to roles that cannot touch production code or design. It is not the independence of separate investigators, and it is not presented as such. Its value is that it removes the most direct route to self-confirmation, not that it supplies an outside party. The arrangement and its lane rules are drawn in Figure 2. The validator lane is shown only partly exercised, because no cognitive claim was ever certified through it: for the three attempts described in Section 4, the validator's gates returned a refusal before any graded number was produced.

3.2 Loop-closure

The second practice governs the inputs a test may use. A test written in the ordinary way hands a stage the input on which it is expected to work, then confirms that the expected output appears.

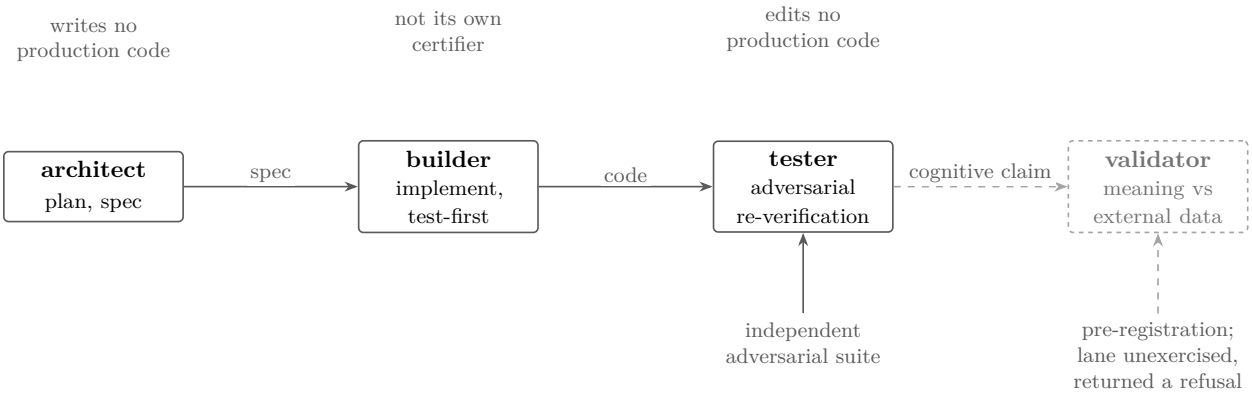


Figure 2: The separation of duties, as roles and contexts operated by AI agents under enforced rules, not independent human teams or external auditors. An architect role specifies and plans; a builder implements test-first; a separate tester adversarially re-verifies each mechanism and never edits production code; a validator grades a cognitive claim against external data under a prediction fixed in advance. The validator lane is shown only partly exercised: no cognitive claim was certified through it, and for the three meaning-bar attempts of Section 4 it returned a refusal before any graded number was computed.

Such a test can pass while the stage is structurally wrong, because it never confronts the stage with anything the stage was not built to handle. It certifies behavior on a fiction of the input, not on the input the system will actually produce. Loop-closure is the rule that each stage must instead be driven by input generated by the stage before it, so that a component is exercised on what the system genuinely feeds it.

This matters because it removes the author's choice of input, which is where self-deception enters. As long as the sources feeding a stage are placed by hand, "it works" means only "it works on my guess of its input," and a stage can carry a structural defect indefinitely with no test going red. Closing the loop, so that perception derives the sources, those sources drive clustering, clustering drives the field, and so on through the cycle, forces each stage to consume the previous stage's real output. It was this substitution of system-generated for hand-placed input that exposed the errors described next, several of which were invisible on friendly inputs and surfaced only once the loop was closed.

3.3 The structural errors the discipline caught

Applied together, adversarial re-verification and loop-closure caught a series of structural errors: defects not in the spelling of the code but in the shape of the algorithm, the kind an ordinary pass on a friendly input slides over. Six are on record. Three are recounted here, and all six are summarized in Table 1. The claim is only that these errors were caught on this system. The list is not offered as exhaustive, and catching them is not evidence that the discipline finds every error or that the code is free of them.

The first concerns the boundary between reading and writing. Roads decay with elapsed time, and the natural implementation computes that decay at the moment a road is accessed and writes the decayed value back. The consequence is easy to overlook: every read becomes a write, and if the same access also re-evaluates whether a road is closed, then a run of reads spread over time can let a closed road drift back open although nothing ever crossed it. A happy-path test passes this by, because reading a value and obtaining the right number is exactly what it checks; the mutation left

behind is invisible unless a test asserts that the stored state is unchanged after the read. The defect was caught by precisely that assertion, a check that a battery of reads leaves the canonical stored state identical, and was repaired by splitting decay into a pure view and a persisting write, with closure re-evaluated only on an active crossing. That split, presented as settled design in Section 2, was not the original design; it is the residue of this error.

The most instructive error lay not in a mechanism but in how a mechanism was judged. An early test of community recovery [Blondel et al., 2008, Traag et al., 2019] reported success: given a graph of two communities joined by a thin bridge, the field partitioned it into the two communities. The success was an artifact of the test. One source had been placed by hand at the centre of each community, so the partition recovered the communities only because the answer had been written into the input. The mechanism was being graded on a fiction of what would feed it, and on that fiction it could not fail. Closing the loop caught it. When the sources were instead derived by perception from real input, the same pipeline produced an incoherent result, its basins collapsing and one source's region spilling across domains. The repair was not to the partitioner but to the validation, which was rebuilt to consume system-derived input, and the original failure was preserved as a standing assertion set to turn green only once the upstream grouping was genuinely fixed. It has since turned green. This is the loop-closure lesson at its sharpest: the error was in the grading, and only system-generated input could expose it.

A third error was carried in from a neighbouring problem. When activation reaches a node along several paths, the contributions must add. An implementation holding the shortest-path intuition, in which the first arrival at a node settles its value, will instead re-expand the node as later paths raise it and propagate the surplus, counting everything downstream of the convergence point more than once. The intuition is correct for shortest paths and wrong for quantities that accumulate. The fix was bounded single-expansion: each node expands once, at the first hop it is reached, carrying the full sum of equal-length paths, and the locking check confirms the downstream value is the exact sum and strictly below what re-expansion would yield.

3.4 Pre-registration and anti-circularity gates

The fourth practice governs claims about meaning, and it is taken directly from the use of pre-registration in empirical science [Nosek et al., 2018]. Before any graded number is computed, the prediction and the direction it must take are fixed and frozen. The artifacts of an experiment, its stimulus lists and selection criteria, are content-hashed before any graded outcome exists, so that what was committed to in advance can be distinguished from what is produced afterward. The selection of stimuli is run blind: it consumes only the structure of the data, never the quantity against which it will later be judged, so the test set cannot be shaped, even unintentionally, by the result it is meant to test.

The rationale is the one pre-registration serves elsewhere. If the prediction may be adjusted after the data are seen, almost any outcome can be dressed as a confirmation, and the test loses its power to discriminate. Fixing the prediction and freezing the design in advance removes that freedom. In this project the gates are not a formality. As Section 4 recounts, on each of the three occasions a cognitive claim was put to them, they returned a refusal before any graded number was computed, because the design could not be populated or graded without circularity.

Taken together, the four practices form one stance toward the system's own claims. Separation of duties removes self-attestation. Loop-closure removes the author's choice of input. The structural errors above are what those two caught when turned on the mechanism. Pre-registration removes the freedom to fit a prediction to the data after the fact. None of this proves that Psyche is correct, that the next error would be caught, or that the discipline transfers beyond this system. What it

Table 1: The structural errors caught on Psyche by adversarial re-verification and loop-closure. Each is a defect in the shape of an algorithm rather than a surface mistake, of the kind a test run on a hand-chosen input would pass. The table records what this discipline caught on this system. It is not offered as exhaustive, and it is not evidence that the discipline finds every error or that the code is free of them. Magnitudes of error are omitted where the present system cannot reproduce them.

Error	Root cause	Fix	Now locked by
Read-as-write	decay computed on access wrote back, so a read mutated state and a closed road could drift open	decay split into a pure view and a persisting write; closure changes only on an active crossing	a battery of reads leaves stored state identical, and time and reads never reopen a closed road
Convergence double-counting	a shortest-path "first arrival is final" intuition applied where activation sums	bounded single-expansion, hop-layered, summing equal-length paths	the post-convergence value is the exact sum, strictly below additive re-expansion
Order-dependent nondeterminism	an activation-ordered frontier, and iteration over a hash-randomized set of identifiers, varied results by process	hop-layered expansion with per-layer frozen values	output is byte-identical across fresh interpreters under randomized process hash seeds
Identity and order leak	tie-breaking on identifiers regenerated each run gave nondeterministic ordering	tie-break on a stable creation index rather than on identity	partition, ordering, and representatives are identical across processes
Binding degeneracy	connected components on a graph connected by construction yield one group; flow-cutting keys on distance, not community	reframe to a multi-source field, with basins around several peaks	a single-seed spread yields several coalitions, and a multi-source field recovers a two-community graph
Circular validation	the answer was placed by hand in the input; a recovery test passed only because a source sat in each community	close the loop so sources are derived by perception from real input	the once-asserted failure on system-derived input has flipped to the passing behavior

supports is the narrower claim this paper makes: on this system, the discipline caught the structural errors recorded here and refused three premature claims, and it did so by construction rather than by chance.

4 The Methodology in Action: Three Meaning-Bar Attempts

The previous section described four practices and claimed, in particular, that the anti-circularity gates refused three cognitive claims before any graded number was computed. This section makes that claim concrete. Three times the project reached for an externally graded cognitive result, and three times the gate declined to grade it. What makes the episode evidence rather than anecdote is that the three refusals were not one obstacle met three times. They were three structurally distinct obstacles: a claim that could not be separated from the construction of the substrate, a claim whose generating mechanism the engine does not contain, and a claim whose external signal does not vary on a realistic workload. A discipline that halts for three different reasons is responding to the structure of each problem, not to a single recurring artifact.

Two points of calibration hold across all three. First, each finding is a statement about what could be established on this substrate with this instrument, not about the phenomenon in the world. The subsections below report that priming could not be separated from the weight map on this corpus, that the fan effect is not produced by the engine as built, and that the chosen interoceptive signal does not vary on a realistic workload. None of them says, or licenses, that priming, the fan effect, or interoception is absent in human cognition. Second, in none of the three cases was a graded result computed. No effect size, no slope, and no direction of effect was produced; the gate fired at selection or at a feasibility check, before the loop was run for score.

4.1 Semantic priming

The first attempt was semantic priming: the expectation that a concept, once activated, leaves related concepts more readily activated than unrelated ones, with relatedness drawn from a published corpus of human word-association norms [De Deyne et al., 2019] rather than from the system. The difficulty is that the corpus encodes association strength as the weight of the edge between two concepts, and Psyche's substrate is built from exactly those weighted edges. A naive priming result would then be uninterpretable. The system could appear to prime simply by reading back the edge weights it was constructed from, with the spreading-activation loop contributing nothing.

The pre-registered design anticipated this and required a weight-blind leverage stratum: a set of prime pairs in which the strong associate and the weak associate carry equal edge weight, matched also for how many edges point at each target, so that any measured difference would have to come from the loop rather than from the weight or from target popularity. The gate was whether that stratum could be populated. A blind selector, consuming only the graph's structure and the published strengths and computing no activation, was run over the full association graph. It found 7 such pairs, against a pre-registered floor of 30. The pre-registered fallback was unreachable for the same reason: the weight-informative pairs so dominate the corpus that the fallback statistic would have had to exceed its own mathematical ceiling of 1 to register an effect. The counts, the dominance ratio, and the size of the graph are given in Appendix B, and the collapse is shown in Figure 3.

The obstacle is not incidental to this corpus; it is intrinsic to the design. The variable the experiment must predict, association strength, is the very quantity the edges encode, so a positive result could always be the weight map read straight back out, which is precisely what the weight-blind stratum exists to prevent. When that stratum cannot be filled, the experiment cannot tell genuine priming-through-the-loop from a mechanical read-back, and running it would settle nothing. The calibrated finding is narrow. On this corpus, with both the edge-weight path and the target-popularity path controlled, priming-through-the-loop cannot be separated from the substrate's own weight map. That is a statement about separability on this corpus, not a claim that priming is absent.

4.2 The fan effect

The second attempt was Anderson's fan effect: the finding that the more associations a concept acquires, the more slowly any one of them is retrieved [Anderson, 1974, Anderson and Reder, 1999]. It was chosen precisely because it escapes the circularity of priming. It rides on the number of a concept's associations rather than on the weight of any one of them, so weight and strength are no longer the same axis. The reason it could not be graded lies one level deeper, in what produces the effect.

In the account the effect comes from [Anderson and Lebiere, 1998], retrieval shares a fixed

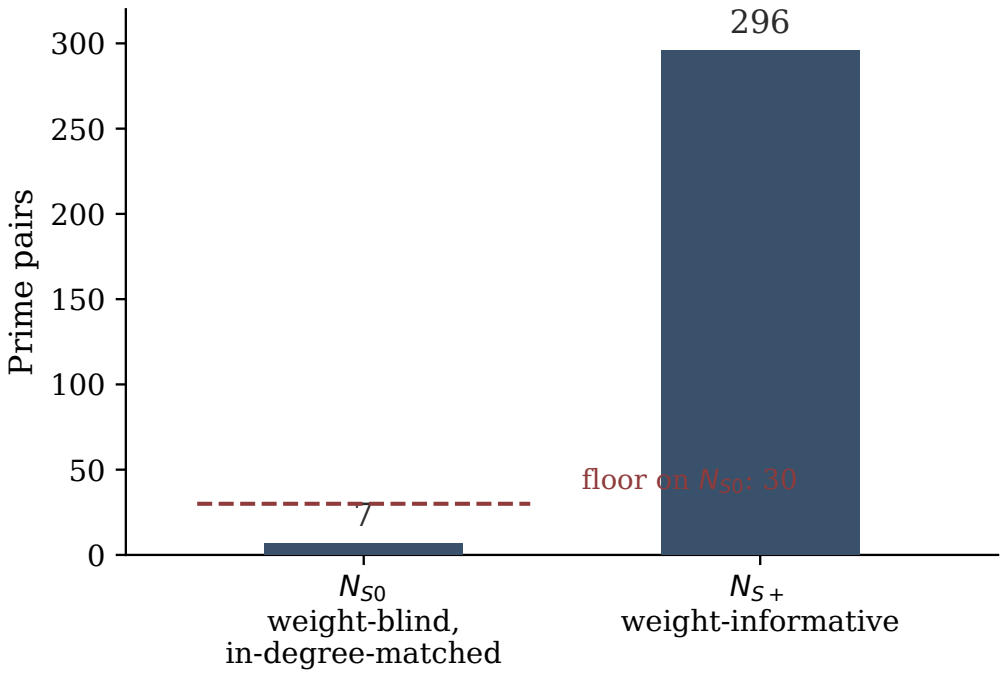


Figure 3: Anatomy of the priming attempt: the weight-blind leverage stratum cannot be populated. Because association strength is encoded as the integer edge weight on a saturating ladder, a prime's strong band almost always outweighs its weak band, so pairs of equal weight are rare. Seven were found against a pre-registered floor of 30, and the fallback statistic would need to exceed its ceiling of 1. No activation and no effect size were computed; the selector consumed only graph structure. The figure shows why priming-through-the-loop cannot be separated from the weight map on this corpus, not that priming is absent.

budget: a source emits a conserved quantity of activation that is divided among its associations, so each receives less as their number grows, and retrieval slows accordingly. Psyche's engine conserves no such budget. Each outgoing road receives the source's full activation, with no division by the number of roads leaving the node. The consequence is direct, and it was measured. As the number of a concept's associations is swept upward, the activation delivered to a target does not fall but stays flat to the last digit across the fan sizes tested. The sweep, identical across those sizes, is reproduced in Appendix B and plotted in Figure 4.

This is a different wall from the first. Priming was circular: a mechanical read-back of the weights would have predicted the right direction, and the gate existed to keep that read-back from being mistaken for a loop effect. Here there is no signal to grade in any direction, because the mechanism that would generate one is not in the engine. The two failures are in a sense opposites: the first had a confound to rule out, the second has nothing to measure. There is an engine change that would create a fan effect, namely dividing a source's activation among its outgoing roads, but it is held as a separate architectural decision, not a repair undertaken to produce the result. Adopting a normalization in order to obtain the effect would not earn the effect, and it would change the meaning of every activation result already verified. The calibrated finding is again narrow. The fan effect is not produced by the engine as Psyche builds it, which says nothing about the fan effect in human memory, a matter not in question here.

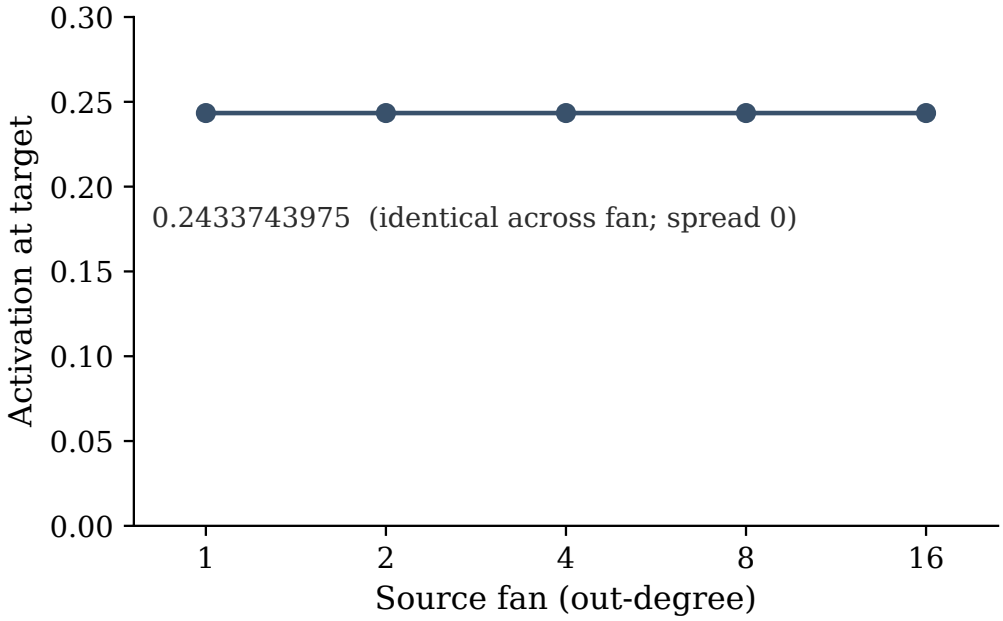


Figure 4: Anatomy of the fan-effect attempt: the activation delivered to a target is flat across fan size. As the number of a concept's associations grows over the sizes tested, the target's activation does not change to the last digit, because the engine gives each outgoing road the source's full activation and divides nothing by the number of associations. The flatness is a property of the engine as built. It is not a claim about the fan effect in human memory. The plot is drawn from the measured sweep in Appendix B.

4.3 Interoception

The third attempt left word association behind entirely; its external signal was the machine itself. The cycle was given a body-state scalar, a measure of friction derived from the gap between elapsed wall-clock time and the CPU time actually consumed, and this friction modulated how far activation was allowed to spread, with higher friction drawing the spread inward, in the direction the cue-utilization account associates with load [Easterbrook, 1959]. The idea follows a lineage in which bodily state biases cognition [Damasio, 1994, Craig, 2002]; here it is tested only as a computational signal. Unlike the first two attempts, the mechanism is verified. A body-state can change which coalition the cycle selects, and a feasibility pilot confirmed that the channel is live, moving the outcome on a large fraction of contested inputs. That fraction is a feasibility measure, recorded as such in Appendix B; it is not a cognitive result.

The obstacle is in the signal, not the channel. Friction defined this way rises only when the process waits off the processor. A realistic Psyche workload does not wait. It is single-threaded, in-memory computation with no input or output to block on, so the CPU time tracks the wall-clock time and the friction sits near zero. The measure is also scale-invariant. Being a ratio, it leaves heavier work to lengthen both clocks together and does not widen as the graph grows. A range wide enough to grade against could be produced only by manufacturing it, for instance by injecting a pause so the process idles, and manufacturing the independent variable is exactly what the anti-manufacture rule forbids. The friction band recorded under a realistic capture, and the fact that its width came from an injected wait rather than from real contention, are given in Appendix B; Figure 5 contrasts the two.

This is the third distinct wall. The signal is not circular, as priming's was, and the modulation

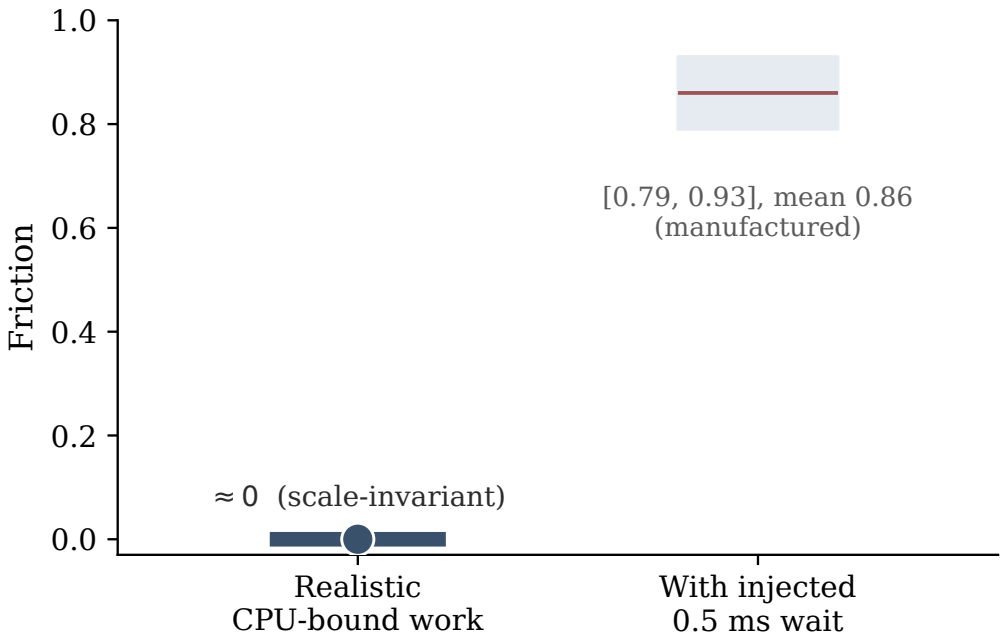


Figure 5: Anatomy of the interoception attempt: the chosen friction signal does not vary on a realistic workload. Friction rises only when the process waits off the processor, but realistic work is CPU-bound and in-memory, so friction stays near zero and, being a ratio, does not widen as the work grows. A gradable range appears only when a wait is injected, which the anti-manufacture rule forbids. The figure concerns the chosen signal, not the modulation channel: the channel is verified and live, and it is the signal that is flat.

mechanism is present, unlike the fan effect's. What is missing is variation: a genuine external signal that the verified channel could read, but which does not move on the work the system actually does. The point bears stating precisely, because it is easy to misread as a defect in the mechanism. The modulation channel is verified and the pilot shows it is live; what is flat is the real-world content of the chosen signal. The contention-tracking claim is therefore unsupported on this substrate with this signal, a statement about what this instrument can measure rather than about interoception. A different body-state signal might vary where this one does not, but it would need its own feasibility pre-check before any graded run, and the obvious candidates carry problems of their own. A memory-pressure signal grows monotonically with the very graph size that drives the competition it would feed, which reintroduces circularity, and a system-load signal is flat on an idle machine and irreproducible under external load.

Three attempts, three refusals, and three different reasons: a claim that could not be separated from the construction of the substrate, a claim whose generating mechanism the engine does not contain, and a claim whose external signal does not vary on the work the system does. In each case the gate fired before the loop was run for score, and in each case the response was to decline to produce a number rather than to manufacture one that could not be interpreted. The substrate and the cycle remained verified mechanisms throughout; what failed was not the machine but the search for an external yardstick it could be graded against without circularity. That the same shape recurred across three unrelated attempts is itself a finding, and it is the subject of Section 5.

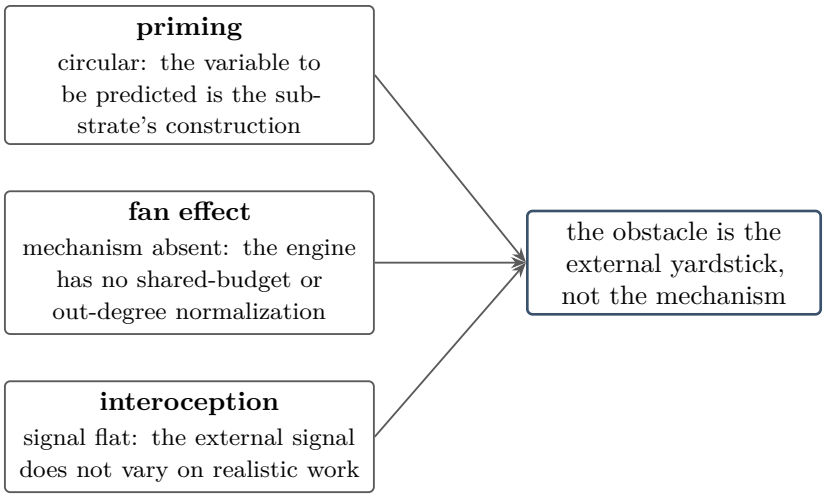


Figure 6: The three attempts converge on one locus. Priming failed by circularity (the variable to be predicted is the substrate's own construction), the fan effect by an absent generating mechanism, and interoception by an external signal that does not vary on realistic work. The three obstacles differ in kind but share a single locus, the join between the mechanism and an external yardstick. The substrate and the cycle remained verified mechanisms throughout. This is a pattern observed on one substrate and these three yardsticks, not a general law about spreading-activation models, cognitive architectures, or cognition.

5 The Meta-Finding

Read singly, each of the three episodes of Section 4 is a local obstacle. Read together, they show a pattern, and the pattern is the subject of this section. In none of the three did the difficulty lie in the mechanism. The substrate computed as specified and the cycle closed as specified throughout. The difficulty lay each time in the relationship between the mechanism and the external yardstick it was to be graded against: whether a signal existed that the built mechanism could both produce and be measured by without circularity. Priming failed because the yardstick was the construction itself, the variable to be predicted being the very weights the substrate is made of. The fan effect failed because the engine does not contain the mechanism that generates the effect, so there was nothing for a yardstick to read. Interoception failed because the chosen external signal does not vary on the work the system actually does. Three different roots, and a single shared locus: not the machine, but the join between the machine and an external measure (Figure 6).

The distinction between a tally and a finding rests on independence. Had the three attempts failed for the same reason, the lesson would be a single nuisance to engineer around. They did not. Circularity, an absent generating mechanism, and a flat external signal are obstacles of different kinds: one is a confound, one is a missing capability, one is a measurement that does not move. That three obstacles so unlike one another arrive at the same locus, the join between mechanism and yardstick, is what makes the recurrence informative. It locates the genuine difficulty in this style of work. Building a substrate that is correct, deterministic, and closes into a loop proved achievable and was achieved. Finding an external measure that such a substrate can both produce and be graded against, without the measure being a restatement of the substrate, is the harder problem.

The scope of this observation has to be stated plainly, because a synthesis invites overreach. What is established is a pattern on one system, Psyche, across three specific yardsticks. It is not a law about spreading-activation models, about cognitive architectures, or about cognition. Nothing

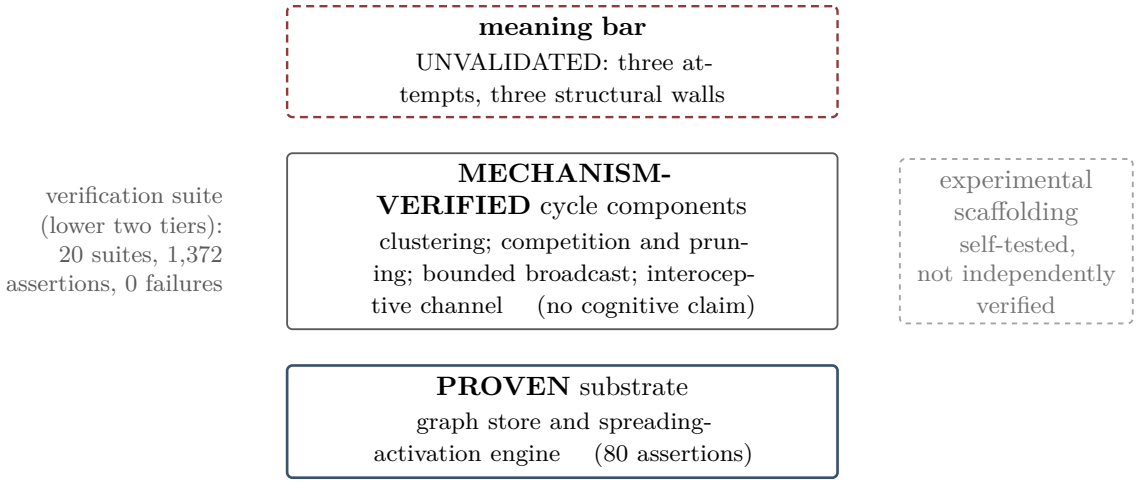


Figure 7: The assurance structure of the work. A proven substrate forms the base, its determinism, integrity, and purity established against hard properties. Above it sit the mechanism-verified components of the cycle, built and adversarially re-verified for the same properties but carrying no cognitive claim. At the top, the meaning bar is left open, marked unvalidated across three attempts and three structural walls. A separate, lower-assurance tier of self-tested experimental scaffolding sits to the side. The verification suite underlying the lower two tiers comprises 20 suites and 1,372 assertions with no failures, detailed in Appendix A; the tiers are kept distinct and are not summed into a single guarantee.

here shows that other substrates face the same join, that other yardsticks would fail in the same way, or that the difficulty is intrinsic to the enterprise rather than to these particular choices. The pattern is real and it was observed three times, which is worth recording. It is not general, and it is not offered as general. A reader is entitled to take from it a hypothesis about where such difficulty tends to sit, not a result about where it must.

It is worth being exact about what did and did not hold. The proven substrate and the mechanism-verified cycle were untouched by any of the three refusals. Their guarantees rest on properties checked independently of any cognitive claim, and no cognitive claim was ever entered against them. What the three episodes record is the discipline operating as designed. The pre-registration and anti-circularity gates exist precisely to stop a claim that cannot be graded without circularity from being graded anyway, and on each occasion they did so before a number was produced. A discipline that halts a result it cannot interpret is not a project failing to obtain one. The two are easy to conflate from the outside, and telling them apart is the entire purpose of the gates: a refused graded run leaves the tier structure intact, whereas a manufactured one would have corrupted it.

The assurance structure that results is shown in Figure 7. A proven substrate forms the base, a layer of mechanism-verified components sits above it, and at the top the meaning bar is left open, marked unvalidated across three attempts and three structural walls. The shape of that figure is the shape of this paper's claims. It also indicates where further effort would have to go. Progress on the meaning bar, for a substrate of this kind, depends less on more mechanism than on the construction of external yardsticks that are gradable and non-circular for a system whose content is its own weighted structure. That construction is not attempted here, and this work does not claim to have solved it. What it offers is a precise account of why the three natural attempts did not, and a map of what stands proven in the meantime.

6 Limitations

The boundaries of this work are best gathered in one place. The first concerns who performed it and how. The work was carried out by one person directing a small set of AI subagents, so the separation of duties on which the verification rests is a separation among roles, not among independent investigators. The adversarial roles, the tester and the validator, operate under enforced lane rules and remove the most direct route to self-confirmation, in which whoever writes a mechanism also certifies it. They do not constitute outside review, and this paper does not present them as such.

It follows that the word verified carries a specific and limited sense here. It means that an independent adversarial role checked a mechanism against hard properties, determinism across processes, integrity, and read-purity, and found them to hold. It is not a formal proof, and it is not a verification of meaning. Where a mechanism carries a known corner case, such as the one in the pruning rule, that case is recorded as a standing negative assertion in the test suite. Recording a gap in this way documents it; it does not close it.

The meaning bar was not cleared. Across the three attempts of Section 4, no cognitive phenomenon was reproduced, and none is claimed. The positive results the paper does make are of two kinds. One is a methodology result: that a particular discipline caught particular errors and declined particular claims. The other is a mechanism result: that a substrate and a cycle hold certain properties. Neither is a cognitive result, and the line between them is held throughout.

The artifact is a prototype, not a system. Its core depends on the Python standard library alone, by design, and every property reported here was established on small graphs. No claim is made about behavior at scale. The one large structure the work touched, the word-association corpus of Section 4, was read only by the blind stimulus selector, which computes no activation; the verified core was never run over it. Nothing here shows that the core would keep its guarantees, or its speed, on inputs of that size. Scale is outside what was tested.

The methodology is demonstrated on a single project, which bounds what may be drawn from it. There was no second project, no control condition, and no comparison against development that omitted the discipline. The evidence is therefore existential: the discipline caught these errors and refused these claims, here. It is not evidence that it would do so elsewhere, nor that it improves on any alternative, and the paper is an experience report that claims only the former. Within the project, three grades of assurance coexist and should not be read as one. The substrate is proven against hard properties. The components of the cycle are mechanism-verified but carry no cognitive claim. A body of experimental scaffolding, self-tested and predating the discipline, sits at a lower grade than either. Only the first two are offered as load-bearing.

7 Related Work

Two distinct lineages meet in this work, one behind the methodology and one behind the artifact to which it was applied. The methodology's testing practice belongs to the tradition of property-based testing, in which a test asserts an invariant over a class of inputs rather than a fixed output for a fixed input [Claessen and Hughes, 2000, MacIver and Hatfield-Dodds, 2019]. The tester's checks are properties of this kind: that a read changes nothing, that a spread is deterministic across processes, that the graph indices stay consistent. A reasonable question follows. Why not simply adopt one of the established property-based testing libraries, several of which are mature and widely used? The answer is a constraint, not a verdict on those tools. The substrate is dependency-free by design and runs on the standard library alone, so the project re-implements a minimal property runner rather than introduce an external dependency. This follows from the dependency-free

constraint and nothing more. The established tools are more capable, and no claim is made that the re-implementation improves on them.

The methodology's second borrowing comes from empirical science rather than from software. The pre-registration of predictions, fixing a hypothesis and the direction it must take before any data are seen, was developed to keep the human sciences from fitting a claim to its result after the fact [Nosek et al., 2018]. Applied here, it governs the grading of cognitive claims about a computational system: predictions and their directions are frozen in advance, and the artifacts of an experiment are content-hashed before any graded outcome exists. The adaptation is itself the point of interest. A discipline meant to keep an experimenter from deceiving themselves about data is turned on a developer's claims about a program they wrote.

The artifact draws on a different body of work, and here the relationship has to be stated carefully, because it is one of design lineage and not of comparison. The shape of the cycle, a competition among coalitions that yields a single winner which is then broadcast, is taken from Global Workspace Theory [Baars, 1988] and from its realization as a computational cognitive cycle [Franklin et al., 2014], with the wider global-workspace literature on selective amplification and internal simulation as further background [Dehaene and Changeux, 2011, Shanahan, 2006, 2010]. The spreading-activation mechanism is taken from ACT-R [Anderson and Lebiere, 1998]. These supply a structural template and a vocabulary. They do not supply a benchmark. Psyche instantiates a control structure of this shape; it is not claimed to instantiate the theories behind it, and it reproduces none of their cognitive results. The fan effect makes the boundary concrete. In ACT-R the effect follows from a shared activation budget divided among a concept's associations [Anderson, 1974, Anderson and Reder, 1999], a mechanism that architecture has and Psyche's engine does not. The consequence, established in Section 4, is that the effect is not produced. That is a difference in what the two engines contain, recorded as a contrast in mechanism, not a comparison of performance between systems.

The substrate's own mechanisms have a parallel and equally loose lineage in neuroscience. The reinforcement of co-active connections during a broadcast recalls Hebbian strengthening [Hebb, 1949], and the saturating bound that keeps the broadcast from diverging echoes homeostatic synaptic scaling [Turrigiano, 2008]. These are analogies of direction, drawn on for design intuition, not claims of biological fidelity.

The interoceptive attempt has a lineage of its own. The idea that a body's physiological state biases what the mind selects has a long history in cognitive science [Damasio, 1994, Craig, 2002], and the particular direction Psyche borrowed, a narrowing of attention under load, comes from the cue-utilization account [Easterbrook, 1959]. As with the cognitive architectures, these are sources of a direction and a rationale, not of a result: the body-state signal was tested only as a computational quantity, and none of the cited findings is reproduced. The question of how a machine might carry an interoceptive variable rather than passively read telemetry has begun to draw contemporary attention [Candia-Rivera, 2026], and persistent runtimes that give a system a form of ambient self-monitoring are being explored in the adjacent, language-model-based setting [Brady, 2026], a different family from the non-LLM substrate studied here. Psyche's contribution to that conversation is the negative structural result of Section 4, not a working interoceptive system.

8 Conclusion

This paper set out to keep two claims apart, that a mechanism is correct and that it means something, while building toward the second. The means was a discipline rather than a result: a separation of duties, loop-closure, adversarial and property-based testing, and pre-registration with anti-circularity

gates, carried through the construction of a cognitive-architecture prototype from the substrate upward. The discipline did concrete work. It caught a sequence of structural errors as the system was built, errors an ordinary pass on a chosen input would have let through, and it refused three premature cognitive claims before any of them became a graded number. What stands at the end is a substrate proven against hard properties and a closed cycle verified as a mechanism.

The deepest lesson is about where the difficulty lay. Building a substrate that is correct, deterministic, and closes into a loop proved achievable, and it was achieved. The harder problem was the one that remained: finding an external yardstick that such a substrate can both produce and be graded against, without the yardstick turning out to be a restatement of the substrate. The three attempts failed at exactly that join, and they failed differently each time. One yardstick was the substrate's own construction. One named a mechanism the engine does not contain. One was a signal that does not move on the work the system does. Three independent failures at one junction, for three unrelated reasons, locate the genuine difficulty in this style of work. It is not the mechanism. It is the relationship between the mechanism and a measure outside it.

The boundary of the work is therefore plain. Psyche is not a mind, and nothing here is a step toward one. No consciousness, no general intelligence, and no validated model of cognition is claimed or implied. The meaning bar was not cleared: across three attempts no cognitive phenomenon was reproduced, and the paper reports that outcome as it stands. What it offers instead are two results of a narrower kind, one about a method and one about a mechanism. The first is that a particular discipline caught what it caught and refused what it refused, on one system. The second is that a substrate and a cycle hold the properties claimed for them. Neither is a cognitive result, and the paper has held that line throughout.

What remains open is the construction of yardsticks that such a substrate can be graded against without circularity, a problem this work locates rather than resolves. The course it commends in the meantime is the one it followed: build what can be verified, claim only what has been, and decline the graded run that cannot yet be interpreted. That is a narrower result than a working cognitive architecture would have been, and a firmer one.

A Verification Summary

This appendix records the mechanism-verification evidence beneath the proven and mechanism-verified tiers. The counts certify mechanism properties, determinism across processes, integrity, and read-purity. They do not certify meaning. Table 2 gives the per-suite assertion counts of the canonical run, grouped by the component each suite exercises. The substrate's two suites contribute 80 assertions; the run totals 20 suites and 1,372 assertions, with no failures. Table 3 records, for each suite that asserts cross-process determinism, the number of randomized process hash seeds under which the result is required to be byte-identical in fresh interpreters.

B The Three Meaning-Bar Attempts: Data

This appendix gives the data behind the three attempts of Section 4. In every case the figures below are those produced before the gate refused a graded run. No effect size, no slope, and no direction of effect was computed for any of the three.

Table 2: The canonical verification suite. Per-suite assertion counts, grouped by the component each suite exercises, read from the project's single test runner. The suite certifies mechanism properties; it does not certify meaning. The apparatus built for the priming attempt is verified here as mechanism, which is distinct from the priming result recorded in Appendix B.

Suite	Assertions
Substrate (proven)	
Graph store	45
Spreading-activation engine	35
Cycle components	
Coalition formation	19
Multi-source field	13
Source clustering	49
Workspace selection	83
Workspace broadcast	63
Perception and end-to-end slice	19
Body-state sensor and modulation hook	49
Independent adversarial re-verification	
Source clustering	73
Source clustering, mutation battery	5
Workspace selection	107
Workspace broadcast	110
Interoceptive channel	350
Priming-attempt apparatus and harness	
Priming selector	67
Priming instrument	42
Priming selector, re-verification	92
Priming instrument, re-verification	76
Priming loop-closure, re-verification	58
Experiment harness	17
Total, 20 suites	1,372
Failures	0

B.1 Semantic priming

The blind selector ran over the full association graph, computing no activation, and populated all four design arms with 300 pairs each. The weight-blind leverage stratum did not populate. Table 4 gives the counts. The global in-degree decile cuts were 1, 1, 1, 1, 1, 2, 2, 5, and 15, and no prime was dropped for an in-degree mismatch. The fallback was unreachable by arithmetic. With the weight-informative pairs dominating, 296 of 300, the weight-only rank-biserial r_W is about 0.97, so the pre-registered margin $r_{MA} - r_W \geq 0.20$ would require r_{MA} to reach about 1.17, above its ceiling of 1. No activation and no graded number were computed.

B.2 The fan effect

The fan sweep held every factor constant except the source's fan and read the activation delivered to a fixed target. Each outgoing road receives the source's full activation, with no division by the number of roads, so the delivered activation does not change with fan. The measured sweep, reproduced from the experiment's output, is the following.

Table 3: Cross-process determinism. For each suite that asserts it, the number of randomized process hash seeds under which the result is required to be byte-identical in fresh interpreters. The activation set is the seeds 0, 1, 3, 7, and 42. The cycle re-verification suites are those for clustering, selection, broadcast, and the interoceptive channel.

Suite	Hash seeds
Spreading-activation engine	5
Coalition formation	8
Multi-source field	8
Perception end-to-end slice	6
Cycle re-verification suites	6

Table 4: Priming: the blind stimulus selection. All quantities are structural counts from the selector, which computed no activation. The stratum count N_{S0} fell below its pre-registered floor, and the fallback was arithmetically unreachable.

Quantity	Value
Association graph, directed edges	1,388,741
Association graph, nodes	133,763
Pairs per design arm (four arms)	300
Weight-blind, in-degree-matched pairs, N_{S0}	7
Pre-registered floor on N_{S0}	30
Weight-informative pairs, N_{S+}	296
Primes dropped for in-degree mismatch	0

F	mean target activation	spread (max-min)
1	0.2433743975	0.0
2	0.2433743975	0.0
4	0.2433743975	0.0
8	0.2433743975	0.0
16	0.2433743975	0.0

The value is identical to its last digit across all five fan sizes, and the spread of activation across targets within each condition is zero. There is no fan-dependent signal to grade.

B.3 Interoception

The modulation channel is verified, and the pilot reported below is a feasibility measure, not a meaning result. Table 5 gives the figures. The pilot drove a deliberately extreme telemetry range, with friction spanning 0.000000 to 0.999202, over 40 contrived, flip-capable percepts; 34 of them changed winner, a fraction of 0.85, above the pre-registered floor of 0.30. This shows that the channel can move the cycle's outcome, which is feasibility, not a cognitive result. The friction band recorded under a realistic capture was narrow and high, 0.79 to 0.93 with a mean near 0.86, but its width came from an injected 0.5 ms yield rather than from real contention. Because friction is a ratio of off-processor time to elapsed time, pure-CPU work of any size leaves it near zero, so a larger graph does not widen the band.

Table 5: Interoception: the feasibility pilot and the friction band. The flip fraction is a feasibility measure on contrived, flip-capable inputs over a deliberately extreme telemetry range; it is not a cognitive result. The realistic-capture band is yield-dominated, set by an injected pause rather than by real contention.

Quantity	Value
Percepts in the feasibility pilot	40
Percepts whose winner flipped	34
Flip fraction (feasibility, not meaning)	0.85
Pre-registered populability floor	0.30
Pilot telemetry range (friction)	0.000000 to 0.999202
Realistic-capture friction band	0.79 to 0.93
Realistic-capture band mean	0.86
Injected yield setting	0.5 ms

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